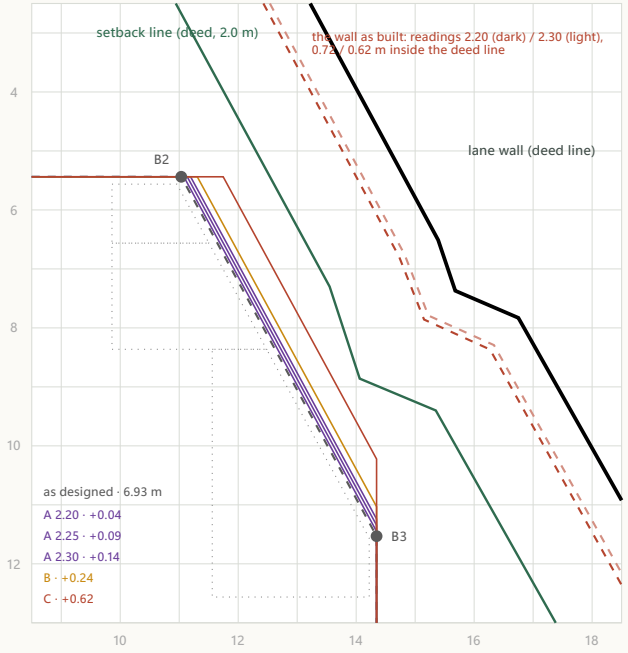


The cut wall enlarged, at the 6 Nov 2026 position. Grey dashed: as designed. Purple: option A for the three possible true readings; amber: B; red: C. Dotted: the rooms along the cut. Where the wall as built really stands is what the owner's precise re-measure at M1 16.25 / D 26.53 (the day-1 point) will fix; the move of A then follows from the table.

Cut wall · shortening it to gain area · short study · APPLIED 6 Nov 2026: −0.12 m (A at the 2.30 reading)

STUDY ONLY · 6 Nov 2026 · start: the house 0.50 m further south (B1 5.49 from M1) · the cut wall 6.93 m

The cut wall B2–B3 is moved outwards, parallel; B2 slides east along the north wall (1.14 d), B3 north along the east wall (2.09 d); every wall sampled at 5 cm. The wall as built: 0.72–0.62 m inside the deed line (readings 2.20–2.30 at the day-1 tightest point, where the model gives 2.92 to the deed line). At the new position the cut wall keeps 0.44–0.54 m from the wall as built and 1.02 m from the deed line as designed.

Option	as designed	A · 2.20	A · 2.25	A · 2.30	B · 0.20 from the wall	C · 0.40 from the deed
The cut wall's move outwards d (m)	**0.00**	**0.04**	**0.09**	**0.14**	**0.24**	**0.62**
New cut length (m) · B2 slides east · B3 slides north	6.93 · 0.00 · 0.00	6.84 · 0.04 · 0.08	6.72 · 0.10 · 0.19	6.60 · 0.16 · 0.29	6.37 · 0.27 · 0.50	5.45 · 0.71 · 1.31
Margin from the deed line (m)	**1.02**	**0.99**	**0.94**	**0.89**	**0.79**	**0.40**
Margin from the wall as built, for a true reading of 2.20 / 2.25 / 2.30 (m)	**0.44** / **0.49** / **0.54**	**0.40** / **0.45** / **0.50**	**0.35** / **0.40** / **0.45**	**0.30** / **0.35** / **0.40**	**0.20** / **0.25** / **0.30**	**−0.19** / **−0.14** / **−0.09**
Gross floor area gained per floor (m²; the roof the same)	**0.00**	**0.27**	**0.61**	**0.94**	**1.59**	**3.86**
Ground floor: the maid room clear (m², now 7.15) · gain · the maid bath · the air tower	7.15 · +0.00 · +0.00 · +0.00	7.29 · +0.14 · +0.08 · +0.04	7.47 · +0.32 · +0.18 · +0.10	7.66 · +0.51 · +0.28 · +0.16	8.02 · +0.87 · +0.49 · +0.27	9.42 · +2.27 · +1.28 · +0.71
First floor: bedroom 1 clear (m², now 13.98) · gain	13.98 · +0.00	14.20 · +0.22	14.49 · +0.51	14.77 · +0.79	15.34 · +1.36	17.54 · +3.55
Bedroom 1: a simple en-suite shower room 1.6 × 2.4 with ≥ 10 m² left (needs 14.34 clear)	no (short by 0.36)	no (short by 0.14)	**yes, by area**	**yes, by area**	**yes, by area**	**yes, by area**
… the hotel layout B4 with its vestibule (needs 15.74 and a rectangle)	no: a wedge	no: a wedge	no: a wedge	no: a wedge	no: a wedge	no: a wedge

What else moves (every option): the three columns on the cut line slide with it (the corner column at B2 east along the north wall by the B2 move, the middle one outwards by d, the column at B3 north along the east wall by the B3 move), the two cut-wall beams and the RC wall W4 shorten, the air tower and the HRV base under the malqaf grow outwards by d over 1.14 m of the cut, the roof's cut edge, the main cornice and the attic's set-back follow the wall; nothing changes in the rooms' doors, windows (the maid's escape casement stays in the cut wall) or the stiffness idea. **B's risk in one line:** 0.20 m leaves nothing for the survey's own tolerance (± 0.10) or the wall's irregularity: one low reading puts the house inside a setback counted from the wall. **C:** valid only if the engineering office confirms in writing that the setback is measured from the deed line; from the wall as built it stands 0.19 m inside the setback.

Verdict and recommendation

1 · With the margin used down to the owner's 0.40 m target (A), the cut wall can move only **0.04–0.14 m**, depending on the precise reading: 0.27–0.94 m² per floor, of which the maid room gains 0.14–0.51 m² and bedroom 1 0.22–0.79 m²; bedroom 1's simple shower room fits by area only from the 2.25 m reading upwards (the hotel layout never).
2 · B (0.20 m) gains 1.59 m² per floor but spends the whole safety margin; C (the deed line) gains 3.86 m² per floor but stands 0.19 m inside a wall-measured setback: neither is a safe decision before the official survey and the office's written answer on where the setback is counted from.
3 · **Recommendation:** re-measure the day-1 point precisely (M1 16.25 / D 26.53, square to the wall's inner face); if it reads ≥ 2.25 m, option A with its move from the table (≈ 0.09–0.14 m, under 1 m² per floor) can go into the drawings after the official survey; if it reads 2.20, keep the cut wall at 6.93 m: 0.27 m² is not worth re-drawing the cut, the columns and the tower. The gain is small in every case; the house's room is in the shift already made.

الخلاصة: بعد إزاحة البيت ٠٫٥٠ م جنوبًا يمكن تحريك جدار القطع للخارج 0.04–0.14 م فقط مع إبقاء ٠٫٤٠ م من الجدار كما هو مبني (بحسب القراءة الدقيقة ٢٫٢٠–٢٫٣٠ م)، أي 0.27–0.94 م² لكل طابق: غرفة الخادمة +0.14–0.51 م² وغرفة النوم ١ +0.22–0.79 م²؛ حمام بسيط لغرفة النوم ١ يتسع بالمساحة من القراءة ٢٫٢٥ فأعلى. الخيار B (٠٫٢٠ م) يستهلك هامش الأمان كله، والخيار C (من خط الصك) يقع داخل الارتداد إن قيس من الجدار. **التوصية:** إعادة القياس الدقيق عند النقطة M1 16.25 / D 26.53؛ إن كانت القراءة ≥ ٢٫٢٥ م فالخيار A بعد المساحة الرسمية، وإلا يبقى جدار القطع ٦٫٩٣ م.

Cut wall study · Study checks passed: the house at the 6 Nov 2026 position (shift 5.563); the day-1 nearest point to the lane line 2.92 m from it (row 18 read 2.20–2.30 there) · every option leaves exactly its wanted margin (within 2 mm): A 0.40 / 0.40 / 0.40, B 0.20, C 0.40 · the cut stays parallel; B1, B4, B5 fixed; the move grows with the reading (A) and with the smaller margin (B, C) · C stands inside a setback counted from the wall as built (−0.19 m at the 2.20 reading): flagged. Sources: base.typ, design_gf.typ (shift 5.563), data/rooms_gf.csv and rooms_ff.csv (the rooms), STRUCTURE_STUDY.md (the cut-line columns), roof.typ (the tower and the cornice), bedroom1_bath.typ (the en-suite rules), SITE_SETOUT.md rows 18 and 23 (the calibration).